

2.701

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P_{\theta_0} \neq P_{\theta_1}$$

①

$$\inf_{\theta_0, \theta_1} \mathbb{P} \left\{ \mathbb{E} \left\{ \mathbb{E} \left(\ell(\hat{\theta}, \theta) \right) \right\} \geq \epsilon \delta \mid \mathbb{P}[\hat{\theta} \neq \theta] \geq \frac{\epsilon}{2} (1 - \text{TV}(P_{\theta_0}, P_{\theta_1})) \right\}$$

$$\text{TV} \leq \sqrt{1 - \rho^2}$$

$$\left. \begin{aligned} B(\theta_0, \theta_1) &= B(\theta_0, \theta_1) \\ B(\theta_0, \theta_1) &= \int_{\theta_0}^{\theta_1} 1 dx = 1 - 2\delta \end{aligned} \right\} \text{TV} \leq \sqrt{1 - (1 - 2\delta)^2} \leq \sqrt{1 - e^{-4\delta n}}$$

$$\mathbb{E} \{ \ell(\theta, \hat{\theta}) \} \leq \frac{\epsilon \delta}{2} \left(1 + \sqrt{1 - e^{-4\delta n}} \right) \rightarrow \delta \geq \frac{\epsilon}{n}$$

$$\inf \sup \mathbb{E} \{ \ell(\theta, \hat{\theta}) \} = O(\delta^2) \rightarrow O\left(\frac{\epsilon}{n^2}\right)$$

② $\hat{\theta} = \min \{ X_1, \dots, X_n \}$

$$F_{\theta}(x) = P_{\theta}(X_{\min} \leq x) = 1 - P_{\theta}(X_{\min} > x) = 1 - P_{\theta}(X_i > x, \forall i)$$

$$= 1 - \prod_{i=1}^n P_{\theta}(X_i > x) = 1 - P_{\theta}(X > x)^n = 1 - (1 - \underbrace{F_{\theta}(x)}_{\text{cdf}})^n$$

$$f_{\theta}(x) = \frac{d}{dx} F_{\theta}(x) = n f_{\theta}(x) (1 - F_{\theta}(x))^{n-1} = \begin{cases} 0 & (x \geq \theta_1) \\ n(1-x+\theta)^{n-1} & (x \leq \theta) \end{cases}$$

$$\forall \theta: \mathbb{E}_{\theta} \left[(1 - \hat{\theta})^2 \right] = \int_0^{\theta_1} (x - \theta)^2 n(1-x+\theta)^{n-1} dx = \int_0^1 u^n (1-u)^2 du$$

$$= n \int_0^1 u^{n+1} - 2u^n + u^{n-1} du = n \left(\frac{1}{n+2} - \frac{2}{n+1} + \frac{1}{n} \right) = \frac{1}{(n+1)(n+2)} \leq \frac{1}{n^2}$$

$$\sup \mathbb{E}(\cdot) \approx O\left(\frac{1}{n^2}\right)$$

1

فیر لکتر مانده این که اگر با یقین را با یقین و توزیع طبعی نه اندر دایره های گشت و گذار است.
که $\alpha(\frac{1}{n})$ است و این کار را نه $O(\frac{1}{n})$ است که کمتر است

سوال 2: اگر P_X و Q_X دو توزیع طبعی باشند

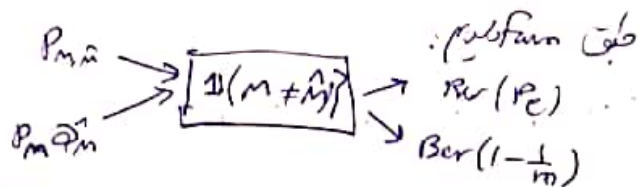
$$I) D_f(P_X T_{A|X}, P_Q T_{A|X}) = D_f(P_X, P_Q) = \mathbb{E}_{P_Q T_{A|X}} \left[D_f \left(\frac{P_X T_{A|X}}{P_Q T_{A|X}} \right) \right] = D_f(P_X, P_Q)$$

$$II) D_f(P_X, P_Q) \geq D_f(P_{\hat{X}}, P_{\hat{Q}})$$

$$X \rightarrow \boxed{T_{A|X}} \rightarrow \hat{X}$$

$$D_f(P_X T_{A|X}, P_Q T_{A|X}) = D_f(P_{X, \hat{X}}, P_{Q, \hat{X}}) \leq D_f(P_{\hat{X}}, P_{\hat{Q}})$$

III



$$\text{Data Processing inequality: } D_f(P_{\hat{X}}, P_{\hat{Q}}) \geq D_f(\text{Ber}(p_c), \text{Ber}(1 - \frac{1}{m}))$$

IV

$$D_f(P_X, P_Q) \geq D_f(\text{Ber}(p_c), \text{Ber}(1 - \frac{1}{m}))$$

$$\inf_{Q_X} D_f(P_X, P_Q) = I(X; \hat{X}) \geq D_f(\text{Ber}(p_c), \text{Ber}(1 - \frac{1}{m}))$$

$$\text{Thus, } \forall Q_X: D_f(P_X, P_Q) = \mathbb{E}_m [D_f(P_{X|m}, Q_X | P_m)] \geq \sum_m \frac{1}{m} D_f(P_{X|m}, Q_X) \leq \frac{1}{m} \sum_m D_f(P_{X|m}, Q_X) = \max_m D_f(P_{X|m}, Q_X)$$

$$\Rightarrow \inf_{Q_X} \max_m D_f(P_{X|m}, Q_X) \geq D_f(\text{Ber}(p_c), \text{Ber}(1 - \frac{1}{m}))$$



for $\theta_0, \theta_1, \theta \in \Theta$; let $\Delta \leq l(\theta_0, \theta), l(\theta_1, \theta)$

$$\tilde{\theta} = \begin{cases} \theta_0 & P = \frac{l(\theta_1, \theta)}{l(\theta_0, \theta) + l(\theta_1, \theta)} \\ \theta_1 & P = \frac{l(\theta_0, \theta)}{l(\theta_0, \theta) + l(\theta_1, \theta)} \end{cases}$$

hence

$$\Rightarrow E_{\theta_0}[l(\tilde{\theta}, \theta_0)] = l(\theta_0, \theta_0) E_{\theta_0}\left[\frac{l(\theta_1, \theta)}{l(\theta_0, \theta) + l(\theta_1, \theta)}\right] \leq \frac{l(\theta_0, \theta_0) E_{\theta_0}[l(\theta_1, \theta)]}{\Delta}$$

$$\frac{l(\theta_0, \theta_0)}{\Delta} E[l(\tilde{\theta}, \theta_0)] \geq E[l(\theta_0, \tilde{\theta})] \geq P[\tilde{\theta} \neq \theta_0] l(\theta_1, \theta_0)$$

since we know that $P[\tilde{\theta} \neq \theta_0] \geq \frac{1}{2} (1 - TV(P_{\theta_0}, P_{\theta_1}))$

we get $P[\tilde{\theta} \neq \theta_0] l(\theta_1, \theta_0) \geq \frac{1}{2} (1 - E_{\theta_0}\{TV(P_{\theta_0}, P_{\theta_1})\})$

which is $E_{\theta_0}\{TV(P_{\theta_0}, P_{\theta_1})\}$

$$\geq \frac{1}{2} [1 - TV\{P_{\theta_0}, E_{\theta_0}[P_{\theta_1}]\}]$$

therefore overall we get that:

$$\Rightarrow E[l(\theta_0, \tilde{\theta})] \geq \frac{\Delta}{2} [1 - TV[P_{\theta_0}, E_{\theta_0}[P_{\theta_1}]]]$$

by Pinsker's inequality that:

furthermore

$$TV(P_1, P_2)^2 \leq \frac{1}{2} D_{KL}(P_1 \| P_2)$$

$$D_{KL}(P_1 \| P_2) \leq D_{KL}(P_1 \| Q)$$

will lead to $TV^2(P_1, Q) \leq \frac{1}{2} D_{KL}(P_1 \| Q)$

thus $C > \frac{1}{2}$

$$\textcircled{\text{III}} \quad 1 + E_2 \left[\frac{|\mathbb{E}[P_0] - Q_X(m)|^2}{Q_X(m)} \right] = 1 + \int_{m \in \mathcal{D}} \frac{E_{\mu_0}[P_0^2(x)] - 2E_{\mu_0}[P_0(x)]Q_X(x) + Q_X(x)^2}{Q_X(m)} dx$$

$$= 1 + \int \frac{E_{\mu_0}[P_0^2(x)]}{Q_X(x)} dx + \int \frac{P_0(x)Q_X(x)}{Q_X(x)} dx + \int Q_X(x) dx$$

$$= 1 + 1 + \int \frac{E_{\mu_0}[P_0^2(x)] E_{\mu_0}[P_0(x)]}{Q_X(x)} dx - 2 \underbrace{E_{\mu_0}[P_0(x)]}_1$$

$$= \int \frac{E_{\mu_0}[P_0^2(x)] E_{\mu_0}[P_0(x)]}{Q_X(x)} dx = E_{\mu_0} \left[\int \frac{P_0(x) P_0'(x)}{Q_X(x)} dx \right]$$

$$\xrightarrow{\text{at limit}} \inf_{\hat{\theta}} E_0[l(\theta, \hat{\theta})] \geq \frac{\Delta}{2} \left(1 - \frac{1}{2} E_0 \left[\int \frac{P_{\theta_1}(x) P_{\theta_2}(x)}{P_{\theta_0}(x)} dx \right] \right)^{1/2}$$

$$TV^2(E_{\mu_0}, P_{\theta_1}, P_{\theta_2}) \leq \frac{1}{2} P_{\mu_0}^2(E_{\mu_0}[P_0], P_{\theta_0}) \leq \frac{1}{2} E \left[\frac{P_{\theta_1}(x) P_{\theta_2}(x)}{P_{\theta_0}(x)} dx \right]^{-1/2}$$



① $X \sim P_0, \theta \in [-\alpha, \alpha]$

$$R_n^*(0) = \inf_{\hat{\theta}} \sup_{\theta \in (-\alpha, \alpha)} E_{\theta}[\|0 - \hat{\theta}\|_2^2] \geq \sup_{\theta \in (-\alpha, \alpha)} \inf_{\hat{\theta}} E_{\theta}[\|0 - \hat{\theta}\|_2^2]$$

$$\geq \inf_{\hat{\theta}} E_{\pi}(\|0 - \hat{\theta}\|_2^2)$$

$\forall \theta: E_{\pi}(\|0 - \hat{\theta}\|_2^2) \geq \text{Var}_{\mathcal{Q}}(\hat{\theta} - 0)$

$\chi^2(P_{\theta, X^n}, Q_{\theta, X^n}) \geq \chi^2(P_{0, \hat{\theta}}, Q_{0, \hat{\theta}}) \xrightarrow{\text{JSS}} \chi^2(P_{0, \hat{\theta}}, Q_{0, \hat{\theta}}) \geq \frac{(E_P(0,0) - E_Q(0, \hat{\theta}))^2}{\text{Var}_{\mathcal{Q}}(\hat{\theta} - 0)}$

$\nexists P_x = Q_x \rightarrow E_P[\hat{\theta}(X^n)] = E_Q[\hat{\theta}(X^n)], E_P[0] = E_Q[0] + \delta$

$R_{\pi}^* \geq \sup \frac{\delta^2}{\chi^2(P_{\theta, X^n}, Q_{\theta, X^n})}$

Assumption: $\chi^2(P_{\theta, X^n}, Q_{\theta, X^n}) = \chi^2(P_0, Q_0) \rightarrow$ which results in using Taylor series

$E_Q[\chi^2(P_{X^n|0}, Q_{X^n|0}) \frac{(dP_0/dQ_0)^2}{q(1)}] \Rightarrow \lim_{\delta \rightarrow 0} \chi^2(P_0, Q_0) = (nI(0) + \alpha(1)) \delta^2$

$\chi^2(P_{X^n|0}, Q_{X^n|0}) = (nI(0) + \alpha(1)) \delta^2$

$I_{\pi} \rightarrow$ Fisher information of π

$\Rightarrow R_{\pi}^* \geq \frac{1}{d^{-2} + nI} = \frac{\alpha^2}{1 + n\alpha^2 I}$

$R_{\pi}^* \geq \frac{1}{I_{\pi} + E_{\pi}(nI(0))} = \frac{1}{I_{\pi} + n\bar{I}}$

$\left. \begin{matrix} \pi' \sim U(-\alpha, 2\delta + \alpha) \\ \pi \sim U(-\alpha, \alpha) \end{matrix} \right\}$

$P_{\chi^2}(P_0, Q_0) = \int_{-\alpha}^{\alpha} \frac{(\frac{1}{2\alpha} - \frac{1}{2\alpha + t\delta})^2}{\frac{1}{2\alpha}} dx$

$= 2\alpha \times 2\alpha \times \left(\frac{2\delta}{2\alpha(2\delta + 2\alpha)} \right)^2 \approx \frac{\delta^2}{\alpha^2} \Rightarrow I_{\pi} = \bar{\alpha}^2$



$$\varepsilon^2 \alpha^2 = \frac{1 - \varepsilon^2}{n\bar{I}} \rightarrow n\bar{I} \alpha^2 \varepsilon^2 = 1 - \varepsilon^2$$

$$\min_{0 < \varepsilon < 1} \max \left(\varepsilon^2 \alpha^2, \frac{1 - \varepsilon^2}{n\bar{I}} \right)$$

boundary $\varepsilon^2 \alpha^2 = \frac{1 - \varepsilon^2}{n\bar{I}} \rightarrow \varepsilon = \sqrt{\frac{1}{1 + n\bar{I} \alpha^2}}$

$$\min_{0 < \varepsilon < 1} \max \left\{ \varepsilon^2 \alpha^2, \frac{1 - \varepsilon^2}{n\bar{I}} \right\} = \varepsilon^2 \alpha^2 = \frac{\alpha^2}{1 + n\bar{I} \alpha^2}$$

Rate distortion $R_n^k(\theta) \geq R_n^* \geq \frac{\alpha^2}{1 + n\bar{I} \alpha^2} = \min_{0 \leq \varepsilon < 1} \max \left\{ \varepsilon^2 \alpha^2, \frac{1 - \varepsilon^2}{n\bar{I}} \right\}$

(II)

$$\frac{\alpha^2}{1 + n\bar{I} \alpha^2} = \frac{1}{\alpha^{-2} + n\bar{I}} = \frac{1}{(\alpha^{-1} + \sqrt{n\bar{I}})^2} \geq \frac{1}{(\alpha^{-1} + \sqrt{n\bar{I}})^2}$$

therefore $\rightarrow R_n^k(\theta) \geq \frac{1}{(\alpha^{-1} + \sqrt{n\bar{I}})^2}$

(III)

wif sup $\hat{\theta} \in [\alpha + \theta, \alpha + \theta]$ $\left\{ \mathbb{E}_{\theta} [\|\hat{\theta} - \theta\|_2^2] \right\} = R_n^k(\theta) \geq \frac{1}{(\alpha^{-1} + \sqrt{n\bar{I}})^2}$

where $\bar{I} = \frac{1}{2\alpha} \int_{\alpha-\theta}^{\alpha+\theta} I(\theta) d\theta$ is the avg of $I(\theta)$ in $(\alpha - \theta, \alpha + \theta)$

$\alpha = n^{-1/4}$, $\bar{I} = \frac{1}{2\alpha} \int_{\alpha-\theta}^{\alpha+\theta} I(\theta) d\theta \xrightarrow{\text{shift, change vars}} \frac{1}{2\alpha} \int_{-\alpha}^{\alpha} I(\theta_0) + o(\alpha) = \mathcal{L}(0,1) + o(n^{-1/4})$

$\Rightarrow$ wif sup $\hat{\theta} \in [\alpha - n^{-1/4}, \alpha + n^{-1/4}]$ $\left\{ \mathbb{E}_{\theta} [\|\hat{\theta} - \theta\|_2^2] \right\} \geq \frac{1}{(\alpha^{-1} + \sqrt{n\bar{I}})^2} = \frac{1}{(n^{1/4} + n^{1/2} \mathcal{L}(0,1) + o(n^{1/4}))^2}$

$\Rightarrow R_n^k(\theta) \geq \frac{o(1)}{n \mathcal{L}(0,1)}$

Q.E.D.